

Intrusion Detection and Prevention Systems

Class 7 — Tuesday, September 15, 2026 · 6:00 pm EDT (Eastern Daylight Time) · LAB_04 starts tonight

1

IDS vs IPS

detect vs block

2

Architecture

network & host

3

Placement

where sensors go

4

CIS Controls

active defense

Fictional client throughout: Laurentide Outfitters inc. — a small Québec company



How tonight runs

0:00–0:04	Recall of Class 6 (host firewalls) → why detection comes next
0:04–0:17	Lesson 1 · IDS and IPS: detecting versus blocking
0:17–0:30	Lesson 2 · Network-based and host-based architecture
0:30–0:42	Lesson 3 · Where to deploy: access points and endpoints
0:42–0:53	Lesson 4 · CIS Critical Security Controls in active defense
0:53–0:58	LAB_04 part 1 briefing (summary — the guide is a separate document)
0:58–1:00	Wrap-up and what to do until 7:30 pm

Graded tonight: nothing. LAB_04 (5%) is graded after part 2 in Class 8. No quiz tonight.



Recall of Class 6 — from firewalls to detection

A firewall decides **WHETHER** traffic may pass

Based on source and destination address, port and protocol.
Last Saturday (LAB_03) every host got a hardening checklist and a short list of inbound firewall rules.

A firewall does **NOT** judge **WHAT** the traffic does

An allowed web request is forwarded even if it carries an attack inside it.
Reading the content of allowed traffic is the job of an IDS / IPS — tonight's subject.

The same allowed packet, two questions:



Goal for the hour: by 7:30 pm you have a sensor installed and watching real traffic in your own isolated lab.



1

Lesson 1

IDS and IPS: detecting versus blocking

Definitions · three detection methods · alert vs drop · what a mistake costs



What an intrusion is, and what an IDS does

An intrusion is any action that violates the security policy of a system or network — including the attempt, not only the success.

IDS — Intrusion Detection System

- Observes traffic or host activity.
- Compares what it sees with its detection logic.
- On a match, writes an ALERT: time, rule, source, destination.
- **Does NOT change the traffic — the packet is already delivered.**
- Called passive and out-of-band: works on a copy, outside the traffic path.

Where the words come from

NIST (National Institute of Standards and Technology) Special Publication 800-94:

“Guide to Intrusion Detection and Prevention Systems” — the reference for tonight's vocabulary.

An ALERT means, in effect: “at this time, this rule matched this event, from this source to this destination.”

csrc.nist.gov/pubs/sp/800/94/final



What “preventing” adds

An IPS (Intrusion Prevention System) = an IDS + a RESPONSE, applied while the packet is still in the device (inline).

Drop

discard the packet silently

Reset

end the session on both sides

Block

refuse the source for a period

One umbrella term: IDPS

NIST uses IDPS (Intrusion Detection and Prevention System) because almost every modern product can run in either mode.

Snort and Suricata — the two free tools of LAB_04 — are IDPS. Tonight you install one as an IDS; in Class 8 you add rules; whether it ever becomes an IPS is a design decision you will take with the table on the next slides.



Detecting vs blocking, stated precisely

Property	IDS — detection mode	IPS — prevention mode
Position	Out-of-band: a copy from a mirror port or tap	Inline: the traffic passes through it
Action on a match	Writes an alert; packet delivered anyway	Alert + drop / reset / block the source
Timing	After delivery — alert can arrive seconds later	Before delivery — each packet is delayed
Cost of a false positive	An analyst wastes time on a harmless event	Legitimate business traffic is blocked
Effect of a device failure	Loss of visibility only; network keeps working	fail-open (traffic flows uninspected) or fail-closed (link cut)

Same detection logic in both columns — only the response and its consequence change.



Four words for every outcome

True positive

alert for a real attack

False positive

alert for something that was NOT an attack

False negative

an attack that produced NO alert

True negative

normal traffic that correctly produced nothing

Tuning = reducing false positives without creating false negatives. Because a false positive is cheap out-of-band and expensive inline, a new sensor is almost always started in IDS mode.



The three detection methods

Signature-based

Match observed data against a list of known patterns (rules).

Precise for known attacks; blind to attacks with no rule yet. Class 8 rule sets are signatures.

Anomaly-based

Build a baseline of normal activity, then alert on strong deviation.

Finds unknown attacks; needs a learning period; unusual-but-legitimate activity causes false positives.

Stateful protocol analysis

Compare observed protocol behaviour with how the protocol should behave.

Catches misuse like a 40,000-character web request; must track each session's state.

Most products combine all three — Suricata matches signatures, parses protocols, and keeps per-session state.



Watch (5 min) + worked example RLE 07-1

► Video — Professor Messer

“Intrusion Prevention”, CompTIA Security+ SY0-701, 3.2 — about 5 minutes.

Detection vs prevention, inline device, mirror-port monitoring, fail-open / fail-closed.

[youtube.com/watch?v=7QuYupuic3Q](https://www.youtube.com/watch?v=7QuYupuic3Q)

RLE 07-1 — one rule set, two modes

Eight web requests reach the shop (10.10.10.20); packets 3, 5, 7 carry a traversal, a SQL-injection and an XSS probe.

Run them through the same three rules as IDS, then as IPS. Same alerts; different delivery.

Result — identical detection, different action:

Mode	Alerts	Delivered to the server	Dropped
IDS (out-of-band)	3	8 → all packets arrive	0
IPS (inline)	3	5 → [1, 2, 4, 6, 8]	3 → [3, 5, 7]

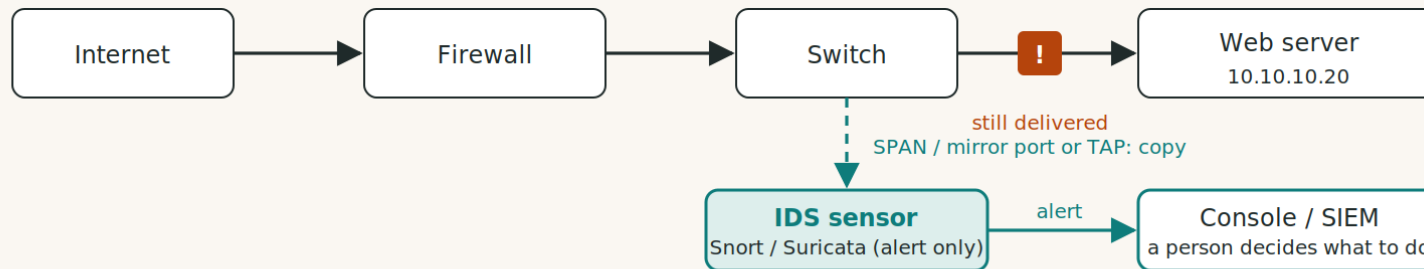
Deliverable: RLE07-1_detection-mode-decision-table.csv — for each zone, the sensor type, the mode to start with, and why.



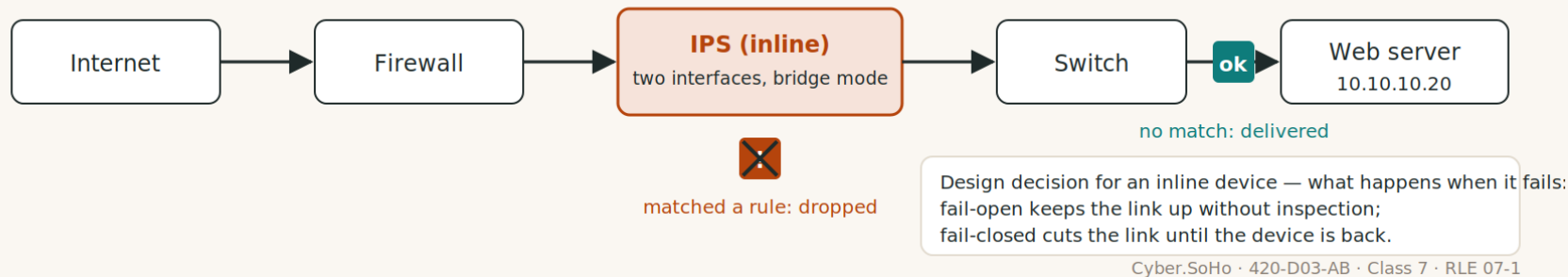
Schematic — out-of-band (IDS) vs inline (IPS)

Two ways to connect the same detection engine to a network

1. IDS — out-of-band: the sensor inspects a COPY of the traffic; the original packet is delivered



2. IPS — inline: every packet passes THROUGH the device and can be dropped before delivery

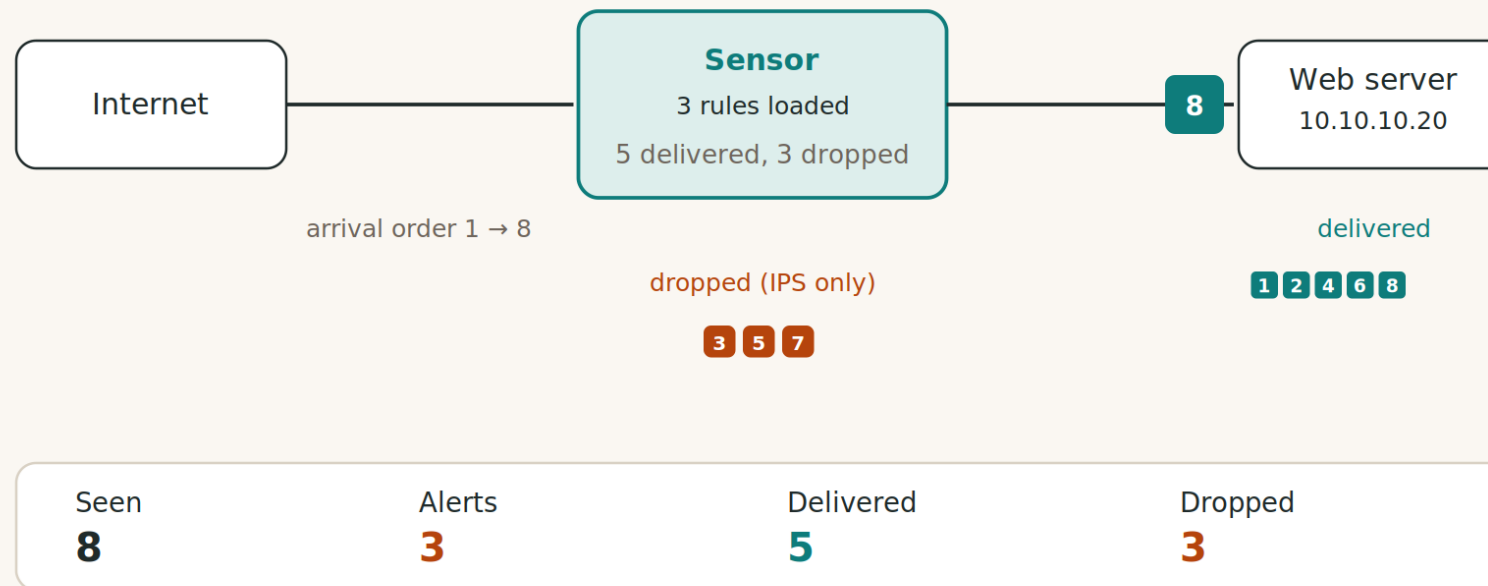


Out-of-band = a copy, no blocking (IDS). Inline = in the path, can block (IPS). Downloadable as SVG and PNG.



What the runner shows — IPS mode

RLE 07-1 · sensor mode: IPS (alert and drop, inline)



Packets 3, 5 and 7 are dropped by the IPS; the same three alerts are written in both modes.



Lesson 1 — remember these

IDS

Alerts and lets the packet continue. Passive, out-of-band, never changes traffic.

IPS

Adds a response. Inline, so it can drop, reset or block before delivery.

False positive

An alert for a non-attack. Costs analyst time in IDS mode; blocks real traffic in IPS mode.

Signature detection

Matches known patterns. Precise for known attacks; blind to attacks with no rule.

Fail-open / fail-closed

What an inline device does on failure: keep passing (availability) or cut the link (security).



2

Lesson 2

Architecture: network-based and host-based

Four components · the sensor pipeline · what a host agent sees · limits of each



Four components, whatever the product

1 Sensor / Agent

Observes. Sensor watches a network segment; agent watches one host.

2 Management server

Collects events from every sensor and agent; correlates them; forwards alerts.

3 Database server

Stores events and alerts for later searching.

4 Console

The interface the analyst uses to read alerts and tune rules.

Tonight in LAB_04 your Snort/Suricata install is a SENSOR, and its text log is the console and database at once — enough for Laurentide Outfitters inc. In Lesson 4 the CIS Controls ask that the alerts of every sensor and agent end up in one place: the management server.



The network sensor pipeline (1/2)

1 · Capture

The network card is set to promiscuous mode and hands the sensor every frame on the wire, not only frames addressed to it.

→ *the interface name you give Suricata; VirtualBox "Promiscuous Mode: Allow All".*

2 · Decode

Headers are read layer by layer: Ethernet, then IP (Internet Protocol), then TCP/UDP, then the application protocol.

→ *the decoder / app-layer sections of suricata.yaml.*

3 · Normalisation

Fragments are reassembled into whole packets and TCP segments into whole streams.

→ *so a rule matches a pattern an attacker split on purpose; the reason a sensor needs memory.*



The network sensor pipeline (2/2)

4 · Detection engine

Rules (signatures), anomaly baselines and protocol checks are applied, grouped by protocol and port.
→ rule files in `/etc/suricata/rules/`; the `HOME_NET` variable = which addresses are “ours”.

5 · Output

Every match becomes an alert: text in `fast.log`, structured JSON in `eve.json`, optionally a packet copy.
→ the files you open with `tail -f` to prove the sensor is running.

6 · Response (IPS only)

When inline, the verdict is applied: pass, drop, or reject.
→ not used in `LAB_04`; the “IPS mode” switch of `OPNsense` / `pfSense` in `LAB_05`.

Two limits follow from the pipeline:

1) it cannot decode an encrypted (TLS) payload — it sees addresses, ports, sizes and the server name, not the content. 2) it sees only the segment it is attached to.



The host-based agent: what it watches

File integrity monitoring (FIM)

Hash baseline of watched files; reports every modified, added or deleted file. One changed byte changes the hash.

Log analysis

Reads OS and application logs; rules like five failed logins in a minute, a new admin account, a stopped service.

Process & registry monitoring

New processes, scheduled tasks, services; on Windows, the registry keys that start programs at boot.

Rootkit detection

Compares what the OS reports with what the agent finds by other means — rootkits hide by altering what the OS reports.

Strength

Works after decryption, inside the host: sees the request received, the file written, the user who did it.

Costs

- Uses host memory & CPU.
- Must be installed on every host.
- An admin-level intruder can stop it.
- Sees nothing of the network.

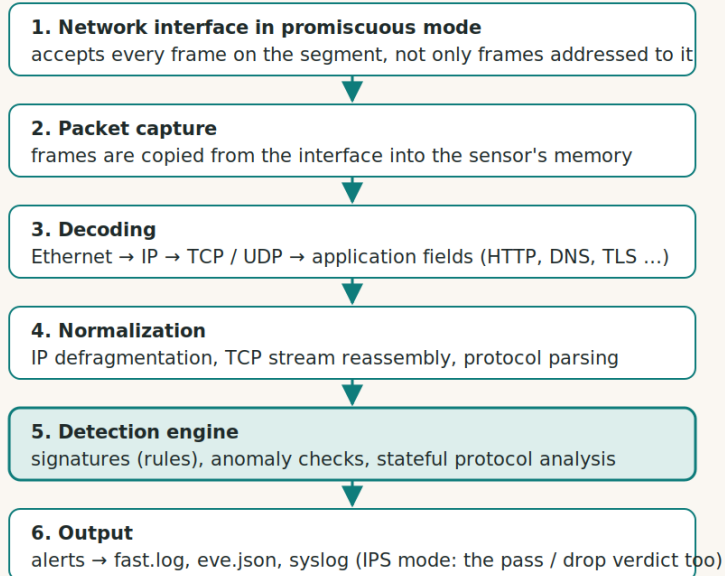
Free tools: Wazuh, OSSEC · on Linux: auditd + AIDE



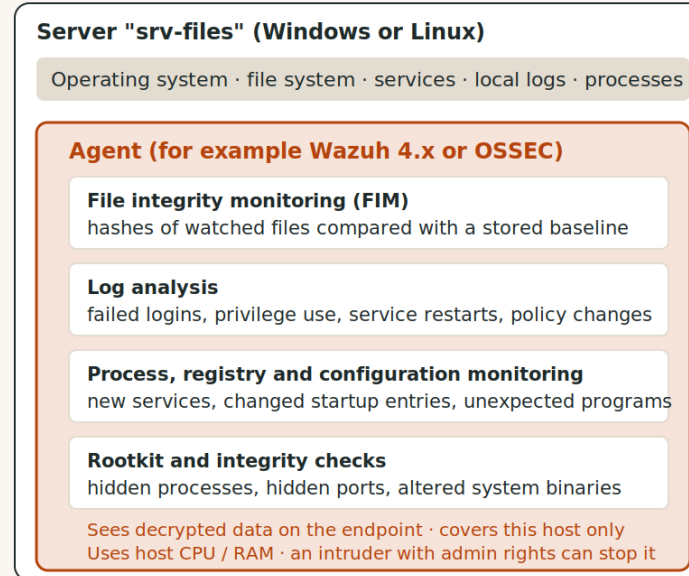
Schematic — anatomy of an IDPS

Inside a network sensor and inside a host agent — the four IDPS components

Network sensor (NIDS / NIPS) — per segment



Host agent (HIDS / HIPS) — per system



Management server + database + console (in the lab: all on one machine, e.g. the Wazuh manager)
receives events from every sensor and agent · stores them · correlates them · shows them to the analyst (CIS Safeguard 13.1)

Cyber.SoHo · 420-D03-AB · Class 7 · RLE 07-2 · component names follow NIST SP 800-94

Network sensor pipeline (top), host agent (FIM, logs, processes, rootkit), and the management server that centralises alerts.



Two more types, and network vs host

WIDS — Wireless IDS

Watches the radio side of Wi-Fi: rogue access points, forced disconnects, unauthorised networks. Reasons come in Class 13.

NBA — Network Behaviour Analysis

Watches flow statistics (who talks to whom, how much) — the natural home of anomaly-based detection.

Criterion	Network (NIDS/NIPS)	Host (HIDS/HIPS)
Installed on	One sensor per segment	Every protected computer
Input	Packets copied from the wire	Files, logs, processes of that host
Sees encrypted traffic	Only metadata, not content	Yes — after the host decrypts it
Sees traffic between other hosts	Yes, if it crosses the segment	No
Can be disabled by an intruder	Only by reaching the sensor	Yes, once admin of the host



Watch (5 min) + worked example RLE 07-2

► Video — Professor Messer

“Intrusion Detection and Prevention Systems”, CompTIA Network+ N10-006, 1.1 — 5:13.

Network-based vs host-based; where the sensor sits; what the host agent sees that the sensor cannot.

[youtube.com/watch?v=mmt4B60xSi0](https://www.youtube.com/watch?v=mmt4B60xSi0)

RLE 07-2 — what the host agent sees

Baseline four files with SHA-256, then detect a change and an addition; scan SSH logs and alert on repeated failures.

A network sensor sees none of this: the session is encrypted and a file edit produces no packet.

What the agent reports that the network sensor cannot:

Finding	How	Network sensor?
payroll_2026.csv MODIFIED	hash changed vs baseline	No — no packet
update.exe ADDED	new file not in baseline	No — no packet
5 failed logins from 20.20.10.45	log analysis, threshold 5	No — SSH encrypted
success after 6 failures — investigate	log analysis pattern	No — SSH encrypted

Deliverable: RLE07-2_fim-baseline.csv — file name, SHA-256 digest, time stamp. Store it on the manager, not on the watched host.



What the runner shows — host vs network view

RLE 07-2 · finished

Network sensor sees

accountant PC
20.20.10.45

TLS

srv-files
SSH, port 22

encrypted session — content not readable

Sessions seen

1

Bytes seen

11,160

File edits seen

0 (not visible)

Failed logins seen

0 (not visible)

What it can report:
"20.20.10.45 opened SSH to srv-files"

What it cannot report:
which file changed · which password failed

Host agent sees (srv-files)

Watched files	SHA-256 (first 16)	status
backup.sh	bd42fd168bed1bc0	unchanged
customers.db	63bc7ffbcdbef50d	unchanged
payroll_2026.csv	d5d781297bff3654	MODIFIED
sshd_config	d65ec1c8aa7f5404	unchanged
update.exe	89d7a811a0ff6621	ADDED

Failed logins from 20.20.10.45

6

threshold 5

20.20.10.31 (technician): 1 failure

ALERT: success after 6 failures — investigate

alerts:

:

Left: the network sensor's blind spots. Right: the host agent sees the file changes and the login pattern.



3

Lesson 3

Where to deploy: access points and endpoints

Five positions · mirror, tap or inline · promiscuous mode · which endpoints first · HOME_NET



One rule decides everything

A network sensor sees only the packets that cross the point where it is attached.

This is how a switched network works — not a limit of any product. A switch sends a frame only to the port of its destination.

Network access point

A place where traffic between two zones must pass:

- the interfaces of the firewall and the uplinks between switches.
- Everything moving from one zone to another is visible there.

Endpoint

A computer itself — a server or a workstation.

- The only way to see what happens inside it is the host agent of Lesson 2.
- This is why servers get an agent, not just a nearby sensor.



Five candidate positions

Position	What it sees	Value	Mode
P1 · Outside the firewall	Everything the Internet sends, incl. what the firewall blocks	Too noisy; never inline	IDS, if at all
P2 · DMZ link (just inside)	Only traffic the firewall allowed in + DMZ→server	Every alert is meaningful; the classic first spot	IDS → IPS later
P3 · User-zone link	Workstations reaching servers and the Internet	Detects infected PCs; some browsing noise	IDS
P4 · Server-zone link	Everything reaching the file server and database	Highest-value for one sensor; data lives here	IDS + host agents
P5 · The firewall itself	All four links at once	No extra machine; OPNsense/pfSense include Suricata	IDS → IPS (LAB_05)

Remember P2: just inside the firewall, fewer alerts and each one is about a packet that reached a company system.



How the copy of the traffic reaches the sensor

Mirror port (SPAN)

The switch copies frames of a port or VLAN to the sensor's port.
Free, quick, out-of-band. But drops mirrored frames first under load.

Network tap (TAP)

A device in the cable sends a full copy of both directions.
Complete copy; a passive tap survives power loss. Costs money; brief cut to insert.

Inline

The sensor bridges two interfaces; every packet passes through it.
The only technique that allows blocking (IPS). Adds delay; failure affects traffic.

Promiscuous virtual interface

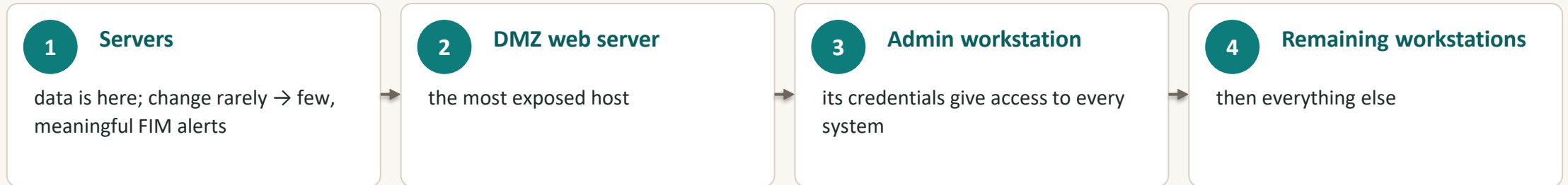
The hypervisor delivers every frame of the virtual network to the sensor.
Free and immediate — what you use tonight. Must be enabled explicitly.

VirtualBox: sensor adapter → Advanced → Promiscuous Mode: Allow All. On AWS/Azure a virtual NIC cannot be promiscuous — use the provider's traffic mirroring, or put the sensor on the same VM as the service.



Endpoints first, and the HOME_NET trap

Which endpoints get an agent, and in which order:



The setting beginners get wrong: HOME_NET

Rules have a direction, e.g. alert http \$EXTERNAL_NET any -> \$HOME_NET any (“alert when a web request goes from outside to one of ours”).

HOME_NET = the address ranges that belong to the company. If it is left at the installation default, an inbound-attack rule never matches, or matches in the wrong direction and alerts on your own outbound traffic.

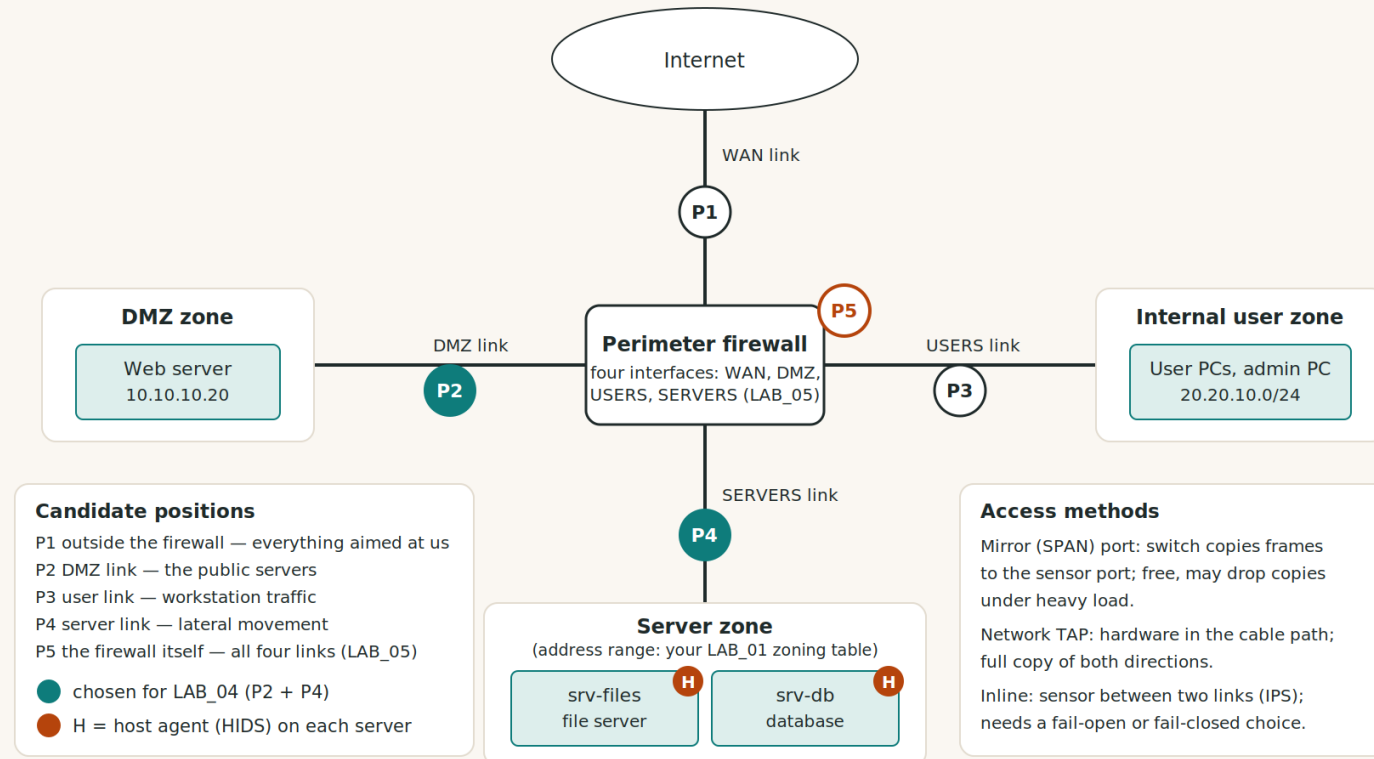
For Laurentide Outfitters inc.: HOME_NET = 10.10.10.0/24 (DMZ), 20.20.10.0/24 (users), and the server range from your LAB_01 zoning table.

It is a required step tonight, and the first thing to check when the sensor “sees traffic but never alerts” in Class 8.



Schematic — candidate sensor positions

Where can a sensor go? Five candidate positions on the zoned network from LAB_01



Cyber.SoHo · 420-D03-AB · Class 7 · RLE 07-3

P2 + P4 tonight (stand-alone sensor); P5 after LAB_05 (firewall runs Suricata). H = a server that also gets a host agent.



Worked example RLE 07-3 — coverage of the network

The task

Seven flows cross the four firewall links; three are marked “must see” (F1 web shop, F2 DMZ→database, F3 users→file server).

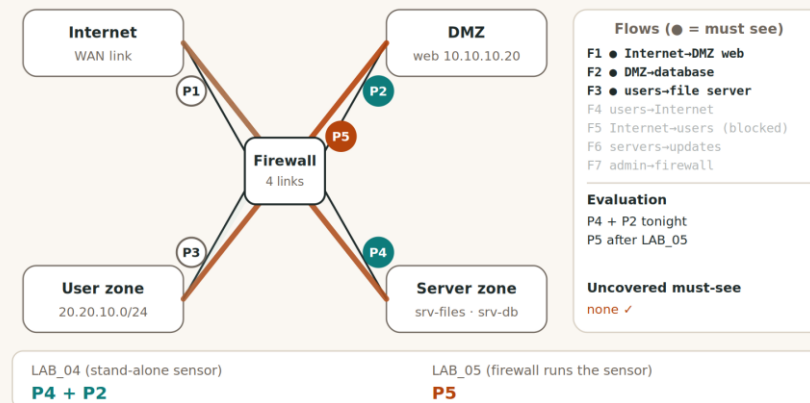
A position sees a flow only if the flow crosses a link it watches.

The result

Stand-alone sensor (tonight): P4 first (covers F2, F3), then P2 (covers F1).

With the firewall (LAB_05): P5 covers all three at once — one position, no mirror port.

RLE 07-3 · finished



Deliverable: RLE07-3_sensor-placement-plan.csv — the rows go into the LAB_05 device registry.



4

Lesson 4

CIS Critical Security Controls in active defense

CIS v8.1 in numbers · active defense · Control 13 · neighbouring controls · Implementation Groups



The CIS Controls in numbers

18

Controls

153

Safeguards

3

Implementation Groups

56

Safeguards in IG1

Each safeguard = one concrete action + one NIST function

A safeguard is written so a technician can answer “done” or “not done”. Its NIST Cybersecurity Framework 2.0 function is one of: Govern, Identify, Protect, Detect, Respond, Recover.

Official pages: [cisecurity.org/controls](https://www.cisecurity.org/controls) · [cisecurity.org/controls/v8-1](https://www.cisecurity.org/controls/v8-1)

VERIFY BEFORE CLASS: confirm the numbers and the IG values against the current CIS Controls v8.1 PDF.



What “active defense” means here

Protect (Classes 1–6)

Zoning, least privilege, hardening, host firewalls.

Configured once, then remain in place. Reduce what an attacker can do — but do not tell anyone an attack is happening.

Active defense (tonight)

Measures that operate DURING an attack: observe, alert, react.

Detect + Respond. IDS = Detect; IPS = Protect (dropping is protective); alert review = Respond, done by a person.

One clarification — the same words, another meaning:

In some publications “active defense” means deceiving attackers with decoy systems, or attacking them back.

Decoys are a legitimate advanced topic. Attacking back is a criminal offence under section 342.1 of the Criminal Code of Canada, whatever the provocation — and is not part of any lesson of this course.



Control 13 — Network Monitoring and Defense (1/2)

#	Title (CIS v8.1)	Where this course does it
13.1	Centralize Security Event Alerting	Lesson 2 (management server); Final Project (Wazuh)
13.2	Deploy a Host-Based Intrusion Detection Solution	Lesson 2; RLE 07-2; Final Project (Wazuh agents)
13.3	Deploy a Network Intrusion Detection Solution	Lessons 1–3; LAB_04 tonight and Class 8
13.4	Perform Traffic Filtering Between Network Segments	Class 2 zoning; LAB_05 firewall rules
13.5	Manage Access Control for Remote Assets	LAB_06 VPN (Classes 16–17)
13.6	Collect Network Traffic Flow Logs	Suricata flow records in eve.json (partly)

13.3 (network IDS) is exactly what you install tonight.



Control 13 — Network Monitoring and Defense (2/2)

#	Title (CIS v8.1)	Where this course does it
13.7	Deploy a Host-Based Intrusion Prevention Solution	Lesson 2 (HIPS); a decision in the RLE 07-1 table
13.8	Deploy a Network Intrusion Prevention Solution	Lesson 1 (IPS mode); OPNsense/pfSense in LAB_05
13.9	Deploy Port-Level Access Control	Class 11 (network device security)
13.10	Perform Application Layer Filtering	Class 8 (web injection); Class 12 (firewall)
13.11	Tune Security Event Alerting Thresholds	Lesson 1 (tuning); Class 8; weekly review (RLE 07-4)

Neighbouring controls a sensor depends on

Control 8 (audit logs: collect 8.2, centralise 8.9, review 8.11) — an alert nobody reads satisfies nothing. Control 12 (secure architecture 12.2) — your zoning table. Control 1 (asset inventory) — the reason HOME_NET can be written. Controls 10 & 17 — handle the malware; have someone assigned to respond.



Why detection comes after hygiene

IG1

Essential cyber hygiene — 56 safeguards, the minimum for every enterprise. Inventory, accounts, patching, audit logs, hardening.

IG2

IG1 + safeguards for an enterprise with IT staff and sensitive data. 13.2 and 13.3 (detection) live here.

IG3

All 153 safeguards. 13.7 and 13.8 (prevention) live here.

Control 13 contains no IG1 safeguard.

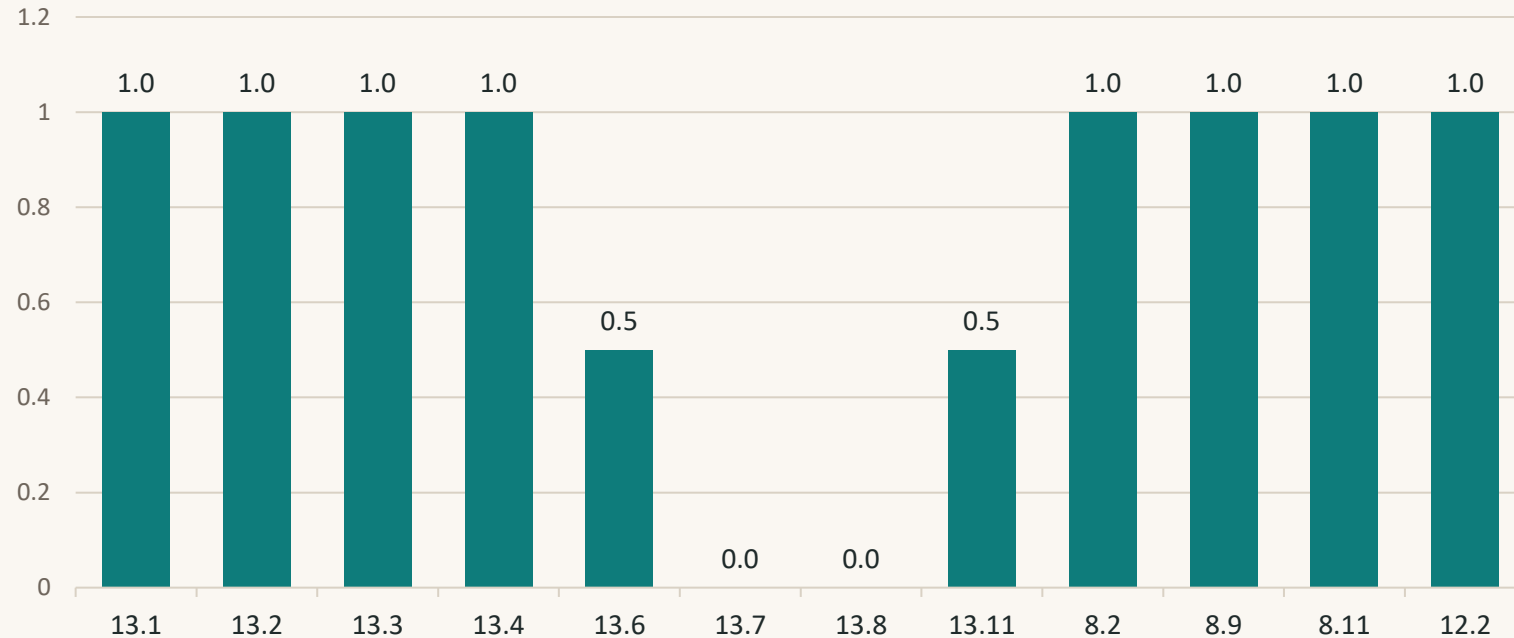
CIS expects the IG1 hygiene (inventory, accounts, patching, the audit logs of 8.2, the hardening you did in LAB_03) to be in place before sensors are deployed — a sensor on an unpatched, uninventoried network produces alerts nobody can act on.

Laurentide Outfitters inc. is IG1 by size but holds customer and payment data, the usual reason a small firm adopts the IG2 detection safeguards. That is the argument you write in the Final Project resilience assessment.



Worked example RLE 07-4 — CIS coverage (native chart)

Coverage of twelve safeguards by the five planned measures (1.0 = fully, 0.5 = partly):



Per control

Control 8: 100%

Control 12: 100%

Control 13: 62.5%

The gaps

- 13.7, 13.8 not covered (no IPS yet).
- 13.6, 13.11 partial.

= the decisions left open in L1–L2.

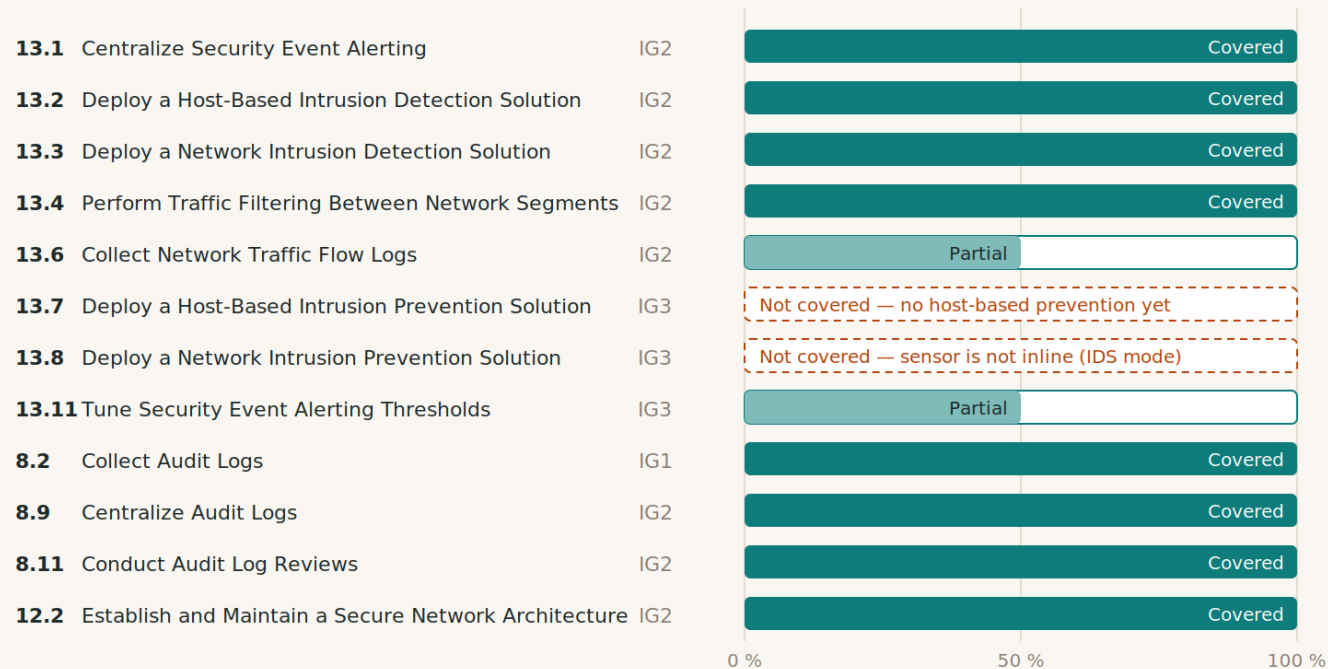
Deliverable: RLE07-4_cis-coverage-report.csv — safeguard, coverage, and the measure behind each row; the evidence table of the Final Project resilience assessment (Class 19).



Schematic — CIS coverage, drawn

CIS Controls v8.1 — coverage of twelve safeguards by the planned detection measures

Bar length = degree of coverage (Covered = full bar, Partial = half bar, Not covered = empty outline). Values from RLE07-4_cis-coverage-report.csv.



Per control: Control 8 = 100 % · Control 12 = 100 % · Control 13 = 62.5 % of the listed safeguards.
The two gaps (13.7 and 13.8) are prevention safeguards: the planned sensors detect, they do not yet block.

Cyber.SoHo · 420-D03-AB · Class 7 · RLE 07-4 · safeguard wording per CIS Controls v8.1 (verify IG column against the current PDF)

Covered (teal), partial (light), not covered (empty). Downloadable as SVG and PNG with the lecture page.



AFTER THE BREAK (7:30 pm)

LAB_04 part 1 — install a sensor

Install one free open-source IDS on a lab machine and prove it sees traffic. The full step-by-step guide is a separate document.

Snort or Suricata — your choice

5% · graded after part 2 in Class 8



LAB_04 tonight — what “done” looks like

“Done” tonight

- One Linux VM runs Snort or Suricata, started with no errors.
- HOME_NET lists the company ranges.
- Packet counters climb while another machine sends traffic.
- Rules and alerts are Class 8 — tonight it only has to run and see.

Common pitfalls

- Sensor on the wrong (management) interface — sees nothing.
- Promiscuous mode left at “Deny” — counters stay at 0.
- HOME_NET left at the default.
- Sensor started twice (service + by hand).
- Too little memory — allocation error.
- No snapshot before installing.

The steps, in the order of the guide:

- 1 Snapshot the Linux VM before installing anything.
- 2 Install the sensor (package source in the guide).
- 3 Set HOME_NET to 10.10.10.0/24, 20.20.10.0/24 and your server range.
- 4 Give the sensor traffic: attach the interface, enable promiscuous mode.
- 5 Fetch a rule set (suricata-update) — you use it in Class 8.
- 6 Start the sensor on the interface; read the start-up log.
- 7 Prove it sees traffic: generate traffic, watch the counters climb.
- 8 Record evidence: 3 screenshots + an environment page.



LAB_04 — the rules that apply to every lab

You may

- Work only inside your isolated lab: VirtualBox, your own free-tier AWS or Azure account, or your CyberQuébec pod.
- Substitute equivalent open-source tools for the suggested ones.
- Build your own helper scripts — they are welcome and belong to you.
- Use AI tools at any stage — if you declare them.

You must never

- Scan, probe, attack, intercept or access any system, network or account that is not yours.
- Create or release malicious software outside the isolated lab.
- Attempt password or interception attacks against any real account or communication.
- Use the material to break any law or policy that applies to you.

AI disclosure rule (academic integrity):

Any use of an AI tool in a submission must state the LLM (Large Language Model) used and the exact prompt(s) given. A local model on 127.0.0.1 is recommended.

The model drafts. The analyst decides. The analyst's name goes on the record.



Wrap-up — the four things from tonight

1

IDS alerts, IPS blocks

Same detection logic; only the response and its consequence differ.

2

Network vs host

A network sensor decodes packets; a host agent reads files, logs and processes.

3

Placement + HOME_NET

A sensor sees only where it is attached — just inside the firewall, and HOME_NET must be right.

4

It is CIS 13.3

Tonight satisfies safeguard 13.3; 13.2, 13.1 and 13.8 wait for later labs.

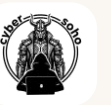
Until 7:30 pm: snapshot → install → HOME_NET → promiscuous mode → start → counters.



Links — standards, documents, videos

NIST SP 800-94 (IDS/IPS guide)	csrc.nist.gov/pubs/sp/800/94/final
CIS Controls (home)	cisecurity.org/controls
CIS Controls v8.1	cisecurity.org/controls/v8-1
CIS Control 13 — Network Monitoring & Defense	cisecurity.org/controls/network-monitoring-and-defense
MITRE ATT&CK	attack.mitre.org
Canadian Centre for Cyber Security	cyber.gc.ca
Criminal Code of Canada (C-46)	laws-lois.justice.gc.ca/eng/acts/C-46/
Video L1 — Professor Messer (SY0-701 3.2)	youtube.com/watch?v=7QuYupuic3Q
Video L2 — Professor Messer (N10-006 1.1)	youtube.com/watch?v=mmt4B60xSj0
Video L3 — Professor Messer (SY0-601 3.3)	youtube.com/watch?v=WPPSsFnWOYg

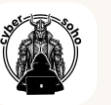
Check every link first: virustotal.com/gui/home/url



Links — the tools of LAB_04 and later labs

Suricata (IDS/IPS)	suricata.io	Snort (IDS/IPS)	snort.org
Emerging Threats Open (rules)	rules.emergingthreats.net	testmynids (optional check)	testmynids.org
Wazuh (host agent)	wazuh.com	OSSEC (host agent)	ossec.net
Zeek (network monitoring)	zeek.org	Security Onion (bundle)	securityonionsolutions.com
OPNsense (firewall, LAB_05)	opnsense.org	pfSense (firewall, LAB_05)	pfsense.org
VirtualBox (lab)	virtualbox.org	VirusTotal URL scanner	virustotal.com/gui/home/url

The CyberQuébec range portal and pod details are on the course platform (VERIFY WITH CYBERQUÉBEC).



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The material shows how to protect systems, not how to attack the systems of others.

Where you may practise

Only inside your isolated lab (VirtualBox, your own free-tier AWS/Azure account, or your CyberQuébec pod), or on systems for which you hold written permission obtained first.

Use snapshots and isolated virtual networks so a mistake stays inside the lab.

Applicable law — unauthorised computer access or use is a criminal offence.

Canada: Criminal Code s.342.1 (unauthorised use of a computer), s.342.2 (possession of a device for it), s.430(1.1) (mischief in relation to computer data). Personal data in the labs is subject to Québec's Law 25 and PIPEDA.

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The instructor never asks for, and never authorises, any illegal act. If an activity seems to touch a system you do not own — stop and ask first.



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Privacy

No real personal data and no real credentials in any deliverable — ever. Screenshots show lab-only accounts and passwords; anything real is redacted before submission (Legal Notice §14).

The fictional payroll and customer records of Laurentide Outfitters inc. stand in for real data.

Artificial intelligence in submissions

Any use of an AI tool must state the LLM used and the exact prompt(s). A local model on 127.0.0.1 keeps data on your machine.

The model drafts. The analyst decides. The analyst's name goes on the record.

Governing law: the Province of Quebec, Canada.



See you after the break — 7:30 pm

The full lecture page has the read-aloud script, the runnable code with a step-by-step runner, the flip-cards and the six questions for each lesson.

Tomorrow, Class 8: write and read rules · safe test traffic · web injection basics · LAB_04 becomes your detection chapter.

